

Chapter 10: Fractions and Proportional Reasoning

Worksheet 10: Fractions and Proportional Reasoning

Learning Objective: Understand and simplify ratios, apply the unitary method, and calculate percentages in real-life comparison and sharing contexts.

Worked Example:

If the ratio of boys to girls in a class of 40 is 3:5, find the number of boys and girls.

Solution: Total parts = $3 + 5 = 8$. Each part = $40 \div 8 = 5$. Boys = $3 \times 5 = 15$; Girls = $5 \times 5 = 25$.

Questions (1–20)

1. Write the ratio 15:25 in simplest form.
2. Find 20% of 150.
3. Express $\frac{3}{4}$ as a percentage.
4. If 5 books cost Rs. 250, find the cost of 8 books using the unitary method.
5. Express the ratio “12 boys to 18 girls” in simplest form.
6. A recipe uses flour and sugar in the ratio 4:1. If you use 2 kg flour, how much sugar is needed?

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7. Find 35% of 240.

 8. A car travels 180 km in 3 hours. How far will it travel in 5 hours at the same speed?

 9. Express the ratio 2:5 as a percentage of the whole (what % the first part is of the total).

 10. A class has 45 students, with boys to girls in ratio 4:5. Find the number of boys and girls.

 11. A shirt marked at Rs. 800 has a 15% discount. Find the discount amount and the selling price.

 12. If 8 workers can build a wall in 12 days, how many days will 6 workers take (assuming the same work rate)?

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13. A map has a scale of 1:50,000. If a distance on the map is 6 cm, find the actual distance in km.
14. A mixture has milk and water in the ratio 5:3. If there are 24 litres of mixture, find the amount of milk and water.
15. In an election, candidate A got 60% of 8,000 votes and candidate B got the rest. Find the votes for each.
16. A photo of 4 cm by 6 cm is enlarged in the same ratio so the longer side becomes 15 cm. Find the scale factor and the new shorter side.
17. A sum of Rs. 3,600 is divided among three friends in the ratio 2:3:4. Find each friend's share.
18. A car covers 15 km per litre, and fuel costs Rs. 102 per litre. Find the cost of fuel for a 450 km journey.

19. Find the ratio of your age to your best friend's or sibling's age. Simplify the ratio.
20. A shop advertises "30% off". If an item's original price is Rs. 1,200, calculate the discounted price.

Reflection Question: Why do you think shops often advertise discounts as percentages rather than fixed rupee amounts? Write your idea.

Real-Life Application Question: Find a recipe, paint-mixing instruction, or juice concentrate label that uses a ratio (e.g. "mix 1 part concentrate with 4 parts water"). Write down the ratio and explain what it means.